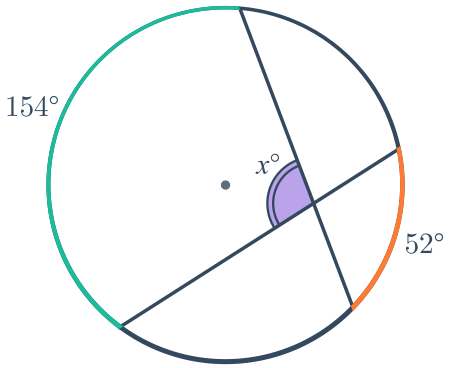
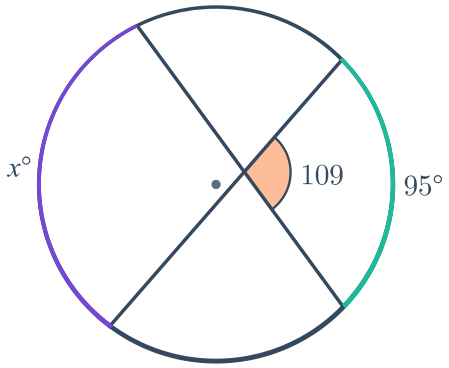
Angles from intersecting chords, secants, and tangents

1 Find the value of the pronumerals for the following figures:

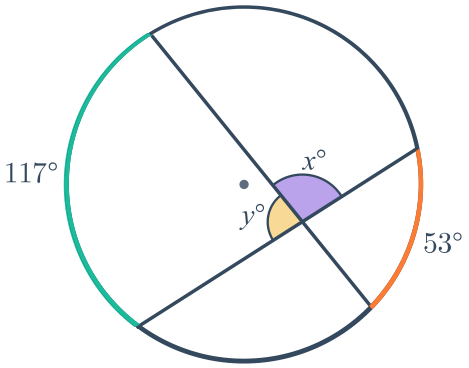
a



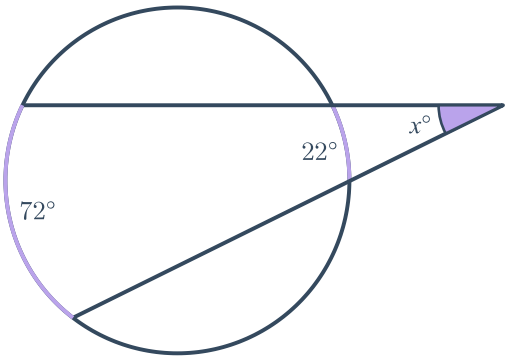
b



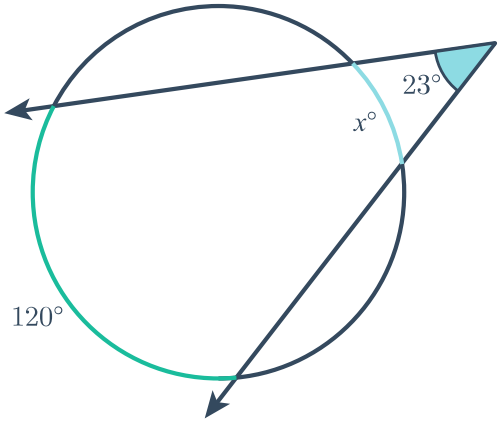
c



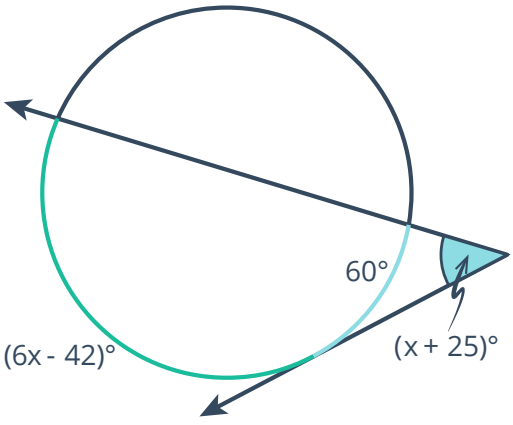
d



e

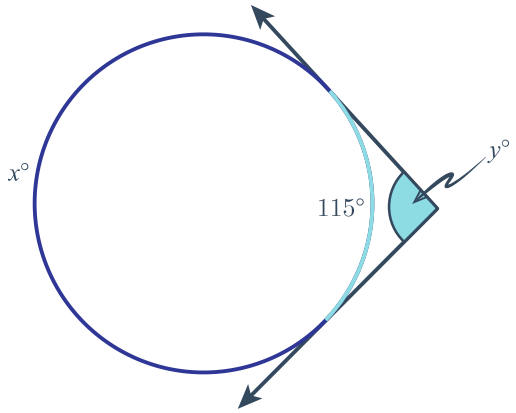


f

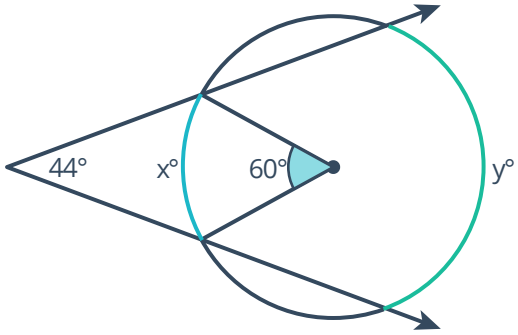


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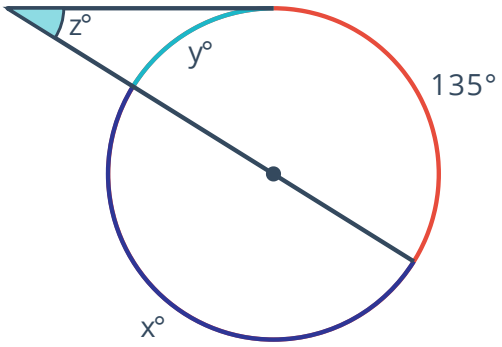
g



h



i

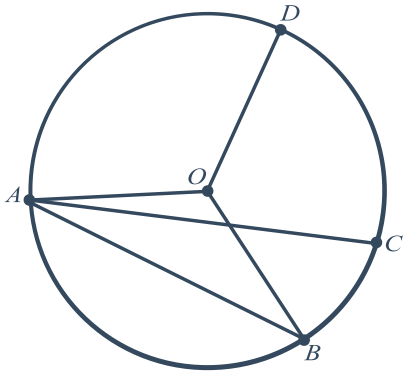


2 In the following diagram, $m\overline{AB} = 8$ and $m\overset{\frown}{AB} = 110^\circ$.

a Find $m\angle AOB$.

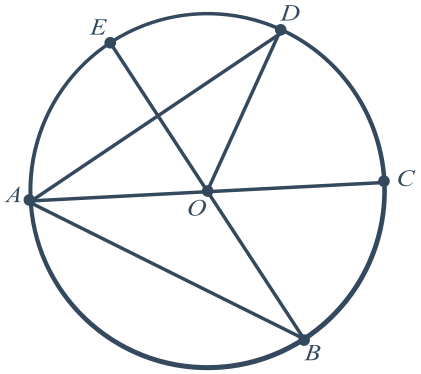
b Find $m\angle OAB$.

c Find the length of the radius, correct to the nearest hundredth.



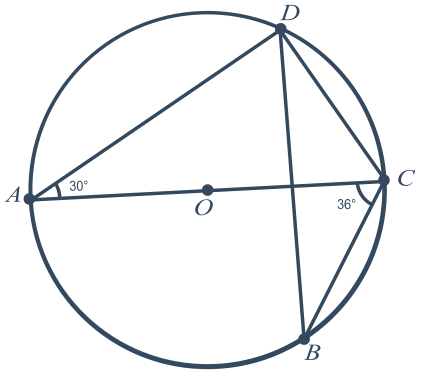
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3 The diagram shows $\overline{AC}$ and $\overline{BE}$ as diameters of the circle and $m\overset{\frown}{BC} = 66^\circ$.



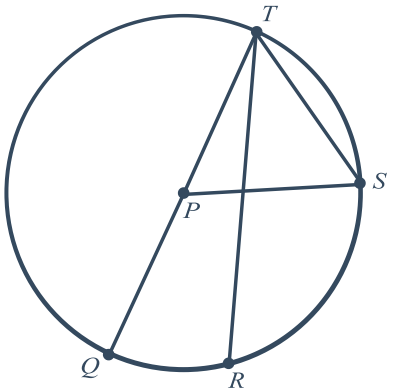
- **a** Find $m\overset{\frown}{AB}$.
- **b** Find $m\angle AOB$.
- **c** Find $m\angle BAC$.
- **d** Find $m\angle ABO$.
- **e** Find $m\overset{\frown}{AE}$.

4 The diagram shows $\overline{AC}$ as a diameter of the circle, $m\overline{AO} = 5, m\angle DAC = 30^\circ$ and $m\angle ACB = 36^\circ$.



- **a** Find $m\angle ADC$.
- **b** Find $m\overline{AC}$.
- **c** Find the exact value of $m\overline{AD}$.
- **d** Find $m\angle ADB$.
- **e** Find $m\overset{\frown}{AB}$.

5 The diagram shows $\overline{TQ}$ as a diameter of the circle, $m\angle TPS = 53^\circ$ and $m\overset{\frown}{QR} = 27^\circ$.



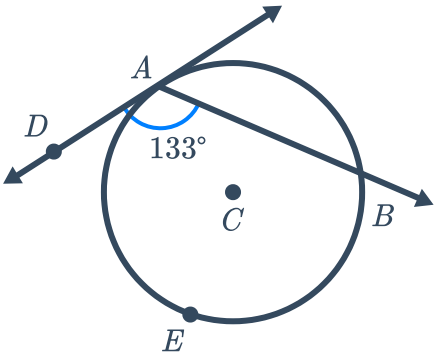
- **a** Find $m\angle QTR$.
- **b** Find $m\overset{\frown}{TS}$.
- **c** Find $m\overset{\frown}{SR}$.
- **d** Find $m\angle RTS$.
- **e** Find $m\angle PST$.

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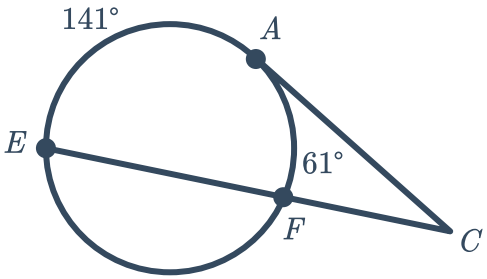
Additional questions

6 For the following diagrams, find the measurement of the indicated angle or arc.

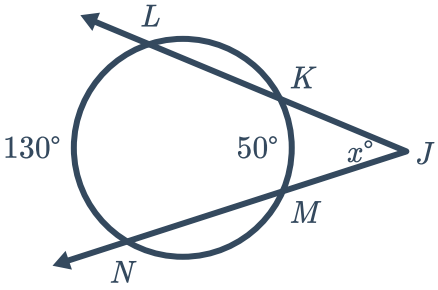
a Solve for $m\overset{\frown}{AEB}$.



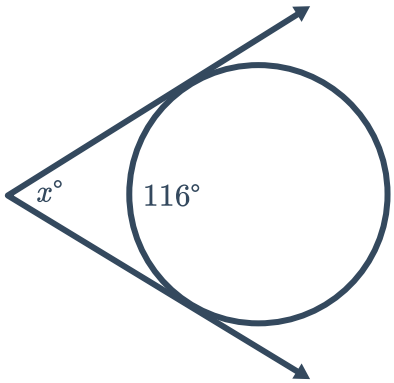
b Solve for $m\angle ACE$.



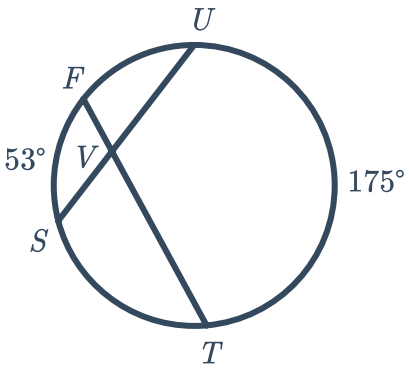
c Solve for x.



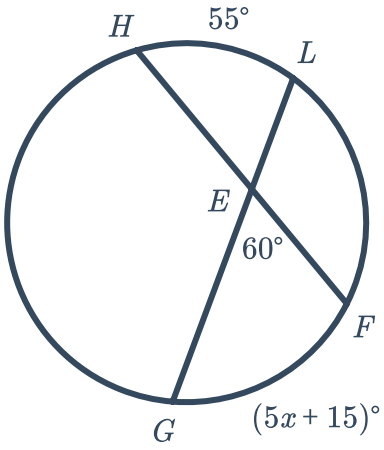
d Solve for x.



e Solve for $m\angle UVT$.

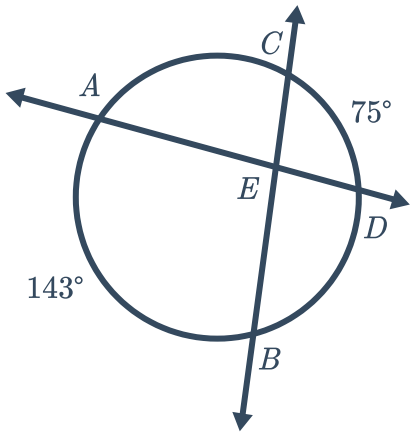


f Solve for x.

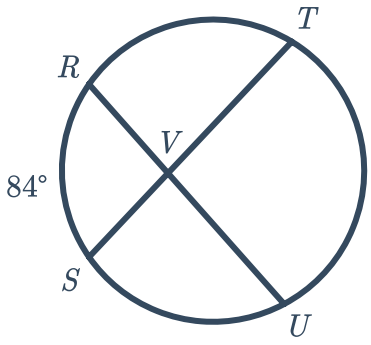


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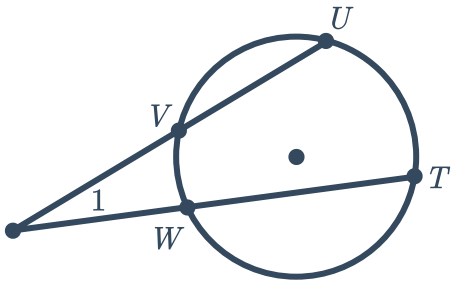
g Solve for $m\angle DEB$.



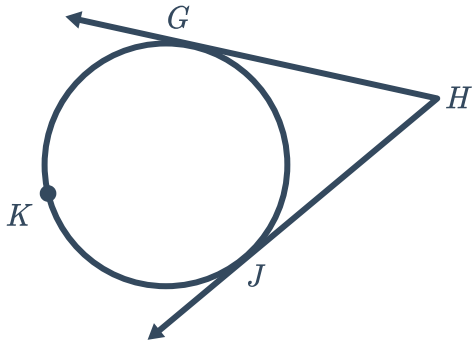
h $m\angle TVU = 107^\circ$. Solve for $m\overset{\frown}{TU}$.



i $m\overset{\frown}{UT} = 110^\circ$ and $m\overset{\frown}{VW} = 24^\circ$. Solve for $m\angle 1$.

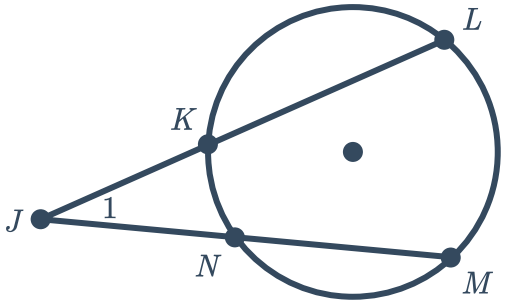


j $m\overset{\frown}{GKJ} = 232^\circ$. Solve for $m\angle GHJ$.



7 Given:

- $m\angle 1 = (3x + 12)^\circ$
- $m\overset{\frown}{LM} = (15x - 5)^\circ$
- $m\overset{\frown}{KN} = (5x)^\circ$



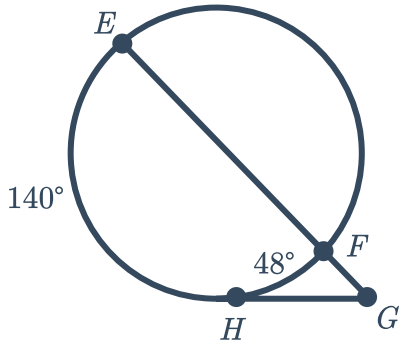
a Solve for x.

b Solve for $m\angle 1$.

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8 Given:

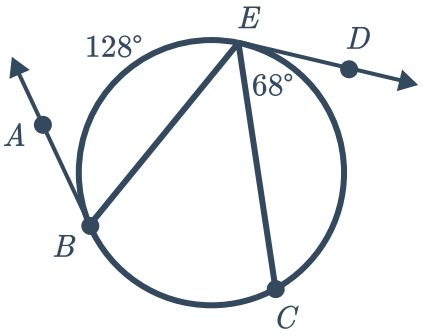
- $m\angle EGH = (5x + 11)^\circ$



a Solve for x.

b Solve for $m\overset{\frown}{EF}$.

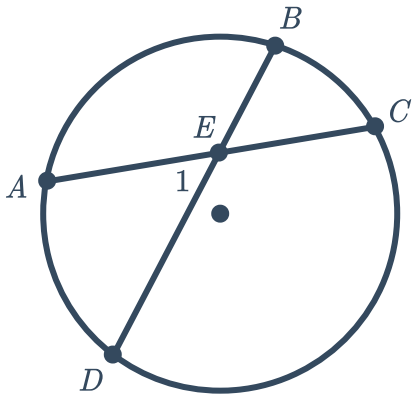
9 Consider the following figures:



a Solve for $m\overset{\frown}{EC}$.

b Solve for $m\angle BEC$.

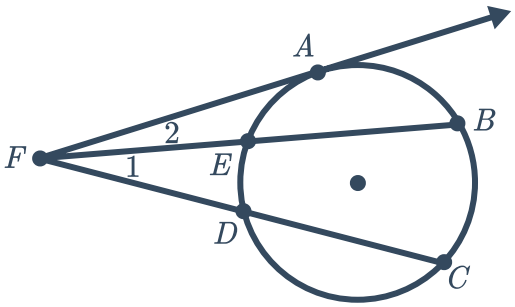
10 If $m\angle 1 = (4x - 8)^\circ$, $m\overset{\frown}{AB} = (10x - 10)^\circ$, and $m\overset{\frown}{CD} = (12x + 26)^\circ$, solve for x.



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11 Given:

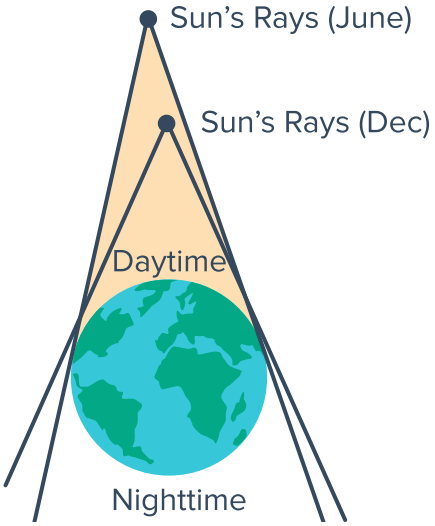
- $m\angle AFD = 75.5^\circ$
- $m\overset{\frown}{BC} = 115^\circ$
- $m\overset{\frown}{AE} = 45^\circ$
- $m\overset{\frown}{ED} = 57.5^\circ$
- $\overrightarrow{FA}$ is tangent to the circle at A.



a Solve for $m\angle 1$.

b Solve for $m\overset{\frown}{AB}$.

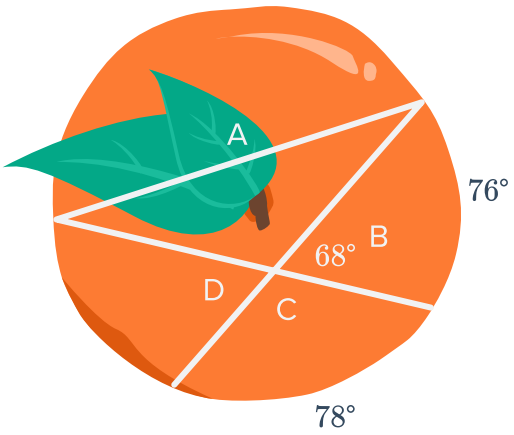
12 The sun's rays hit the Earth such that the tangent rays determine when daytime and night time are, and the time and Earth's rotation determine when certain locations have sun. As the seasons change throughout the year, the amount of daylight changes as well.



a If the arc representing daytime sun exposure in December has a measure of 120°, what is the angle formed by the sun's tangent rays?

b If the arc representing daytime sun exposure in June has a measure of 178°, what is the angle formed by the sun's tangent rays?

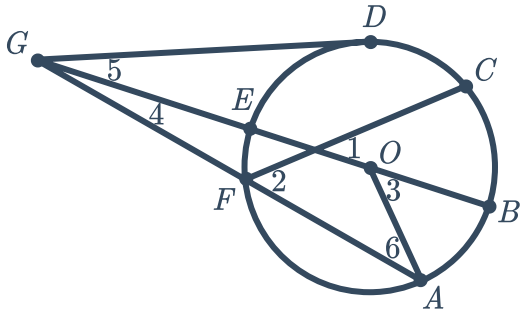
13 Benton is slicing peaches to make his famous peach gelato. In order to fit through the opening on his food processor, the arc on each slice must not exceed 140°.
If Benton made the cuts shown, which pieces would fit through the food processor?



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14 Given:

- $m\angle AGB = 12^\circ$
- $m\overset{\frown}{AB} = (2x)^\circ$
- $m\overset{\frown}{EF} = 44^\circ$
- $m\overset{\frown}{CB} = (3x - 47)^\circ$
- $\overline{EB}$ is a diameter of circle O.



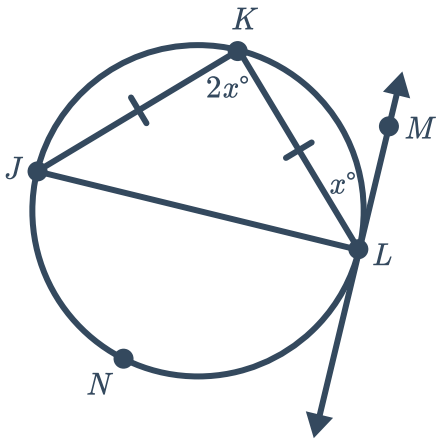
a Solve for x.

b Solve for $m\angle 1$.

c Solve for $m\overset{\frown}{EDC}$.

d Solve for $m\angle 2$.

15 $\overleftrightarrow{ML}$ is tangent to the circle at L.
Show that $\overline{JL}$ is a diameter of the circle.



16

a Draw circle J with $\angle KLM$ formed by two secants, and whose vertex lies outside of the circle and an inscribed angle with the same intercepted arc. Draw an inscribed angle whose intercepted arc is the same as that of $\angle KLM$. Label the vertex of this new angle N.

b Describe the relationship between $\angle KLM$ and $\angle N$.

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